River Erosion Detection Methods: Classical to Modern

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The paper "From Pixels to People" (Rafat et al., 2025) traces river erosion detection through eight groups of methods, from field surveys to fine-tuned foundation models. The groups below follow that order, with the studies the paper cites for each. Numbers in brackets are the paper's reference numbers.

1. Field-based and manual methods

These methods measure the bank directly, on the ground or from low altitude. They are the most accurate, but cover only short reaches. The Pixels paper calls them slow and expensive [6, 7]. The list of techniques below comes from the Mekong study's review (Binh et al. 2020).

Field surveys

1. **Fix reference points.** Benchmarks or monuments are set back from the bank so every survey measures from the same place.
2. **Measure the bank.** Typical tools:
   - Total station or GNSS cross-sections across the bank top and toe
   - Echo-sounder bathymetry for the underwater slope
   - Erosion pins: steel rods driven into the bank face, where the exposed length shows how much soil was lost
   - Terrestrial photogrammetry or laser scanning
3. **Repeat the survey** after each flood season, or after a major event.
4. **Compute change.** Bank retreat (m) along each cross-section, or eroded volume from the difference between two surveyed surfaces.

Used for: calibrating and validating other methods, and designing bank protection at a specific site.

Limits: labour-intensive and slow to repeat. Field crews cannot reach active banks safely during the monsoon, when most erosion happens.

Drone and LiDAR surveys

1. Fly a drone or aircraft over the reach before and after the flood season.
2. Build a digital elevation model (DEM) from each flight, using LiDAR or photogrammetry (structure from motion).
3. Subtract the two DEMs (a "DEM of difference") to get eroded area and volume.

An Australian study used repeat LiDAR plus aerial photos to map bank failures across a whole catchment after an extreme flood (Grove et al. 2013).

Limits: expensive. LiDAR does not see through water, so the underwater bank needs separate bathymetry.

Manual interpretation of satellite images

1. Load images of the same reach from two dates.
2. An analyst traces the bank by eye on each image.
3. Overlay the lines; the area between them is erosion or accretion.

Limits: depends on cloud-free images, and results vary with the analyst. This is the step the GIS methods in section 2 formalise and later methods automate.

Relevance to a predictor: these methods produce the most trusted ground truth, but only for small areas. Use any you can obtain as a validation set, not as training data.

2. GIS and remote-sensing bankline change analysis

This is the standard way erosion has been measured on Bangladesh's big rivers since the 1970s Landsat archive began. Bank lines are drawn on images from several years, then compared in GIS to get eroded area and retreat rates.

1. **Choose images.** Landsat (30 m, from 1972) or later SPOT and Sentinel-2. Dry-season images (November–March) are preferred: they are cloud-free and the water is near its normal level.
2. **Pre-process.** Georeference all images to one coordinate system and correct for atmosphere, so the dates line up pixel for pixel.
3. **Separate water from land.** Either classify the image (supervised or unsupervised), threshold a water index such as NDWI or MNDWI, or digitise the bank line by hand.

4. **Extract the bank line.** Convert the water edge to a vector line for each date. On braided rivers like the Jamuna, the outer bank is usually used, and sandbars (chars) are tracked separately.
5. **Overlay and measure.** Two common ways:
   - **Polygon overlay:** land in year 1 that is water in year 2 = erosion; the reverse = accretion. Gives areas in km² or ha.
   - **Transects:** cast lines at fixed spacing perpendicular to a baseline and measure how far the bank moved along each, giving metres per year. The USGS Digital Shoreline Analysis System (DSAS) automates this. A Jamuna study using transects found the central reach retreating up to 128.5 m/yr between 1990 and 2020 (Islam & Mitra 2024).
6. **Summarise.** Tabulate erosion by bank, reach, district or decade, and map hotspots.

Studies of this kind that the paper cites:

Study	River	Finding
Hassan et al. 2017 [14]	Jamuna	East and west bank erosion and accretion, 1995–2015
Rahman & Islam 2017 [15]	Padma (Harirampur)	Course moved about 12 km north over 150 years
Bhuiyan et al. 2017 [16]	Padma	Left bank eroded 189.4 km², 1973–2011
Billah 2018 [17]	Padma	About 50,000 ha lost in four decades
Islam et al. 2009 [19]	Padma, Jamuna	Bank erosion and channel movement
Chu 2006 [18]	Yellow River delta	Non-Bangladesh example
Kummu 2008 [20]	Mekong (Vientiane)	Non-Bangladesh example
CEGIS 2009 [3]	Three main rivers	Government erosion forecasting and warning tools

Limits the paper lists: manual delineation brings human bias, the method needs cloud-free images, and multi-date analysis is computationally heavy.

Operational use in Bangladesh: CEGIS has run an erosion prediction for the Jamuna, Ganges and Padma since 2004 (CEGIS). Each year it checks the previous year's prediction against early-year satellite images. Its yearly reports list vulnerable land and buildings by district (2026 report).

Relevance to a predictor: this workflow is how you would build your own labels: yearly bank lines, then retreat per transect or per segment. Transect-based retreat (m/yr) works well as a regression target.

3. Improved remote-sensing techniques

All three turn satellite images into a land/water map for each date, then call land that became water "eroded". They differ in the sensor, the rule that separates land from water, and how the bank line is drawn.

Radar (Sentinel-1 SAR) — Freihardt & Frey 2023, Jamuna [7]

This method maps where farmland and houses were lost in each monsoon, about one month after the monsoon ends. The official optical assessment has to wait for cloud-free images, which arrive between November and January. (paper)

1. **Get the images.** Sentinel-1 ground-range-detected scenes in Google Earth Engine: wide-swath mode, 10 m pixels, 30–45° incidence, VV polarisation (VH only exists from 2017). Earth Engine holds amplitude only, so the method uses backscatter strength, not radar phase.
2. **Reduce speckle.** Radar images are grainy. Averaging more dates keeps detail but delays the result; spatial filtering is fast but blurs. They tested 2 weeks to 7 months and filters from 3×3 to 50×50 pixels. One month with a 3×3 boxcar filter was the best compromise.
3. **Classify land cover by backscatter thresholds.** Water is darkest, then sand, fields and trees. They sampled 10 patches of 100 × 100 m per class along the river. Each threshold sits midway between the two neighbouring classes' value ranges (mean ± n standard deviations). Trees are capped at −2 dB so settlements are not counted as trees.
4. **Apply the erosion rule.** A pixel is eroded if it was field or trees before the monsoon and sand or water after it. The pre-monsoon map averages November–April; the post-monsoon map can use November alone. The key sand/water vs vegetation threshold is −13.2 dB with 6 months of images and −12.7 dB with 1 month.

5. **Detect houses as persistent scatterers.** Houses reflect radar steadily over time; vegetation does not. A pixel is a settlement if its amplitude dispersion (standard deviation ÷ mean) is at most 0.4 and its backscatter is at least −4 dB. Both orbit directions are used, because roofs face different ways. A house is eroded if it sits on land that became sand or water.
6. **Check accuracy.** There was no field ground truth. The radar map was compared with Sentinel-2 NDVI (below 0.1 = sand/water, above = vegetation) for November and March over 3 years, giving over 87% agreement.

Output: an Earth Engine tool with eroded-land and eroded-settlement layers for each monsoon, 2015–2019, alongside CEGIS's predicted hotspots.

Limits: fields and trees are confused about 16% of the time (harmless for erosion), many houses under trees are missed, and radar is not combined with optical data.

Relevance to a predictor: it gives yearly event labels. The authors also observed that erosion recurs where it happened in previous seasons, which suggests past erosion is a useful predictor feature.

Water and bank detection indices — SRBED, Mekong 2020 [22]

This method draws an accurate bank line from 30 m Landsat pixels, then measures how far the line moved between two dates. (paper)

Note: the Pixels paper cites this as "Trung et al. 2020". The published first author is Doan Van Binh (Nguyen Xuan Tung is a co-author); the DOI is 10.3390/rs12203298.

1. **Pick comparable images.** Landsat Level-2 surface reflectance, clouds masked. Images are taken in the dry season at matching water levels, because floodwater over the banks would inflate the river and water level shifts the visible edge.
2. **Compute water indices.** Five band ratios that make water stand out: NDVI, NDWI, MNDWI, WNDWI and AWEInsh.
3. **Split land from water.** Three ways were compared: a fixed threshold per index, Otsu's automatic threshold, and a morphology-based method (M-AMERL). With the indices, this made 11 combinations.
4. **Trace the bank with M-AMERL.** Compute how sharply the index changes from pixel to pixel (Sobel gradient). Ignore pixels with no pure water within 2 pixels. Use Otsu on the remaining gradients to find bank pixels, then a watershed algorithm draws one continuous bank line. This copes with fish ponds beside the river, which fooled plain thresholds.

5. **Refine to sub-pixel.** A bank pixel is part water, part land. Its value is modelled as α × water + (1 − α) × land, using pure pixels in the surrounding 5 × 5 window. The line is shifted inside the pixel by the water fraction α (spectral mixture analysis, SMA).
6. **Measure change.** Cut the earlier bank into 200 m segments from fixed control points. Fix two inland reference points per segment, measure the area from them to each year's bank line, and subtract. A smaller area means erosion; a larger one, deposition.

Validation: bank lines hand-digitised from Google Earth for 2014 and 2019, with total error under 7 m.

Results: NDWI + M-AMERL + SMA was best. SMA cut error by about 20%, and NDVI was the worst index, about 70% worse. On 100 km of the Tien River (2008–2014): net erosion 1.5 km², mean bank retreat 2.64 m, over 60 m in places.

Relevance to Bangladesh: turbid water and ponds near the bank are common here too. Matching water levels between dates matters even more on the braided Jamuna.

Colour-channel thresholding — Rafat et al. 2025 (this paper)

This is the quickest option: a rough land/water split from the image's colour values, used only to tell SAM where to look.

1. **Rough split.** A simple analysis of the image's colour channels separates water from land.
2. **Turn it into a prompt.** That rough mask becomes the colour-mask prompt for SAM (the alternative is SAM's automatic prompt).
3. **Segment with SAM.** SAM predicts candidate masks; the one with the best predicted IoU is kept, cleaned with morphological operations, and its pixels counted for area.

The paper does not say which channels or threshold values it uses. Colour alone is unreliable on sediment-laden water, which is also where zero-shot SAM failed, so this step is a starting point that the fine-tuned model corrects.

4. Susceptibility and risk modelling

Instead of detecting erosion that already happened, this method scores every location by how likely it is to erode. It combines several mapped factors with weights set by experts.

Fuzzy multi-criteria geospatial expert system — Alam & Ahamed 2023, Jamuna [21]

(paper abstract)

1. **Map the factors.** Build one raster per factor from Landsat 8 OLI images (2020) and elevation data. The Pixels paper names vegetation indices, slope and elevation among them.
2. **Standardise with fuzzy membership.** Each factor is rescaled to 0–1, meaning "how strongly this value points to vulnerability". Fuzzy curves give a gradual score instead of a hard cut-off.
3. **Weight the factors with AHP.** Experts compare the factors in pairs (for example, "is slope more important than vegetation?"). The analytical hierarchy process turns those judgements into a weight per factor.
4. **Overlay.** A weighted sum of the standardised layers in GIS gives a vulnerability score per pixel, then classes such as high, moderate, marginal and not vulnerable.
5. **Verify.** The authors checked the map against 150 reference points of the Jamuna's main channel taken from 2020 Google Earth Pro images.

Result: the abstract reports high-to-moderately vulnerable areas falling between 3 and 7 km, and recommends land-use planning that accounts for those zones.

Limits: the weights come from expert opinion, not from data, so the map is hard to validate. It is static (one year), with no timing of when erosion will happen.

Relevance to a predictor: the factor layers are a ready list of candidate features. A machine-learning model can learn the weights from your observed erosion labels instead of expert pairwise judgements.

5. Classical machine learning

Here a model learns patterns from data instead of following hand-set rules. The paper cites two kinds.

Self-Organizing Maps (SOM) on LiDAR and spectral imagery — Daneshmand 2013 [23]

A SOM is an unsupervised neural network. It arranges similar input samples next to each other on a small 2-D grid of nodes, so they fall into natural clusters without needing labels.

1. **Build a feature vector per pixel or bank point.** For example LiDAR elevation, slope and roughness, plus spectral bands or indices from imagery, for two or more dates.
2. **Train the SOM.** Each input is matched to its most similar node, and that node and its neighbours are nudged towards the input. Over many passes the grid organises into clusters.
3. **Label the clusters.** An analyst names clusters, for example stable bank, actively eroding bank, deposition, water.
4. **Estimate erosion.** Map the eroding clusters and measure their change between dates.

The paper cites this as a ResearchGate item, and I could not find the full text, so the steps above describe the general SOM approach, not that study's exact setup. A 2024 study shows a related use: SOM to group inputs, then an LSTM to predict bank movement and channel width (Huang 2024).

Limits: clusters need a person to interpret them, and LiDAR is rarely available for Bangladesh's rivers.

Artificial neural network forecast — Rahman et al. 2024, Jamuna at Ulipur [24]

The Pixels paper lists this as "Sarker et al."; the chapter's authors are Md Naimur Rahman, Md Mushfiqus Saleheen, Hamza El Fadili and Md Nazirul Islam Sarker (chapter).

1. **Map past erosion.** Classify Landsat images for several years (supervised and unsupervised classification) and extract erosion and accretion between dates (abstract).
2. **Train an ANN** on the historical erosion and accretion series.
3. **Project forward.** The trained network extrapolates erosion and accretion to future years.

Result: eroded area grew from 3,101 ha (2003–2013) to 4,232 ha (2013–2022). The model projects a sharp decline, to 132 ha by 2042.

Limits: it extrapolates a trend over decades without river flow or other drivers as inputs, so it cannot respond to an unusual monsoon. The projected decline runs against the recent increase and is untested.

Relevance to a predictor: this is the only true forecasting study the paper cites. It shows the basic setup (past erosion in, future erosion out), and why adding drivers such as discharge matters.

6. Deep learning segmentation (CNNs)

Convolutional neural networks (CNNs) learn to label every pixel, for example water or land, from examples. They replace hand-picked thresholds with features learned from data.

How a U-Net is used for erosion mapping

A U-Net has two halves. The encoder shrinks the image step by step to recognise what is in it. The decoder grows it back to full size to say exactly where, with skip connections that carry fine detail across.

1. **Prepare training data.** Cut satellite images into tiles (for example 256 × 256 pixels) and pair each tile with a hand-drawn mask (water, land, sometimes sandbar or settlement).
2. **Train.** The network predicts a mask for each tile, and a loss such as cross-entropy or Dice measures the mismatch with the true mask. Weights are adjusted over many passes. Augmentation (flips, rotations, brightness) stretches a small dataset.
3. **Predict.** Run the trained model over images from each date to get water/land masks.
4. **Detect erosion.** Compare masks between dates: land that became water is erosion (the same rule as section 7).

DeepLabV3+ works the same way but uses dilated convolutions to see wider context at once, which helps with large water bodies.

A second use: predicting the next mask directly. A 2025 study trained a U-Net on the Global Surface Water dataset for the Brahmaputra–Jamuna. Given past years' water maps, it predicts the next year's, and its 2019 predictions of erosion and deposition areas were compared with what happened (EGU25 abstract). This is close to the predictor you are building.

Limits: needs many labelled tiles; struggles with sediment-laden water that looks like sand; a model trained on one river or sensor may not transfer.

Most CNN examples the Pixels paper cites come from neighbouring tasks, not river erosion itself:

- Coastline and shoreline detection [25, Nguyen 2022]
- Building detection from drone images for riverbank monitoring [26, Boonpook 2018, over 90% accuracy]
- Lake water-body extraction [27, Liu 2024]
- WaterNet, an ensemble built for water segmentation [28, Erdem 2021]

- U-Net and DeepLabV3+ for river water segmentation, which a fine-tuned SAM beat [30, Moghimi 2024]

7. Foundation models: the paper's own approach

SAM (Segment Anything Model, Meta 2023) is a general-purpose segmentation model pre-trained on over a billion masks. The paper adapts it to erosion with very little training.

How SAM is built

- **Image encoder:** a large vision transformer (ViT-H, 632M parameters) turns the image into a 64 × 64 grid of features.
- **Prompt encoder:** takes hints about what to segment: points, boxes or a rough mask.
- **Mask decoder:** a small transformer (about 4M parameters) that combines the two and outputs three candidate masks with a predicted IoU score each.

Step 1: zero-shot test

The authors first tried SAM unchanged, with automatic, point, box and mask prompts. It found clean land–water edges but failed in sediment-laden transition zones and could not handle change over time. That is why they fine-tuned it.

Step 2: build the dataset

1. 500 Google Earth Pro images (2003–2025) of erosion-prone sites such as Mokterer Char, Kedarpur and Chowhali, under 10% cloud, mostly dry season.
2. Exported at 1308 × 764 px, normalised to about 10 m per pixel, histogram-equalised, then resized to 1024 × 1024 for SAM.
3. Polygons for water, stable land and erosion drawn in CVAT, reviewed twice (annotator agreement kappa 0.87). Erosion is only 9.2% of pixels, a strong class imbalance.
4. Split 250 train / 50 validation / 200 test. Training images augmented with flips, rotations, elastic deformation and brightness changes.

Step 3: fine-tune only the decoder

1. Freeze the image encoder; train only the mask decoder (about 0.6% of parameters).
2. Loss = 20 × Focal + Dice + IoU. Focal loss focuses on hard pixels, which helps the rare erosion class.

3. AdamW, learning rate 1×10^{-4} with cosine decay, batch 4 (16 with accumulation), mixed precision, on one 16 GB GPU. Training stopped early after about 11 epochs.

Step 4: detect and measure erosion

1. For each date: resize and normalise, encode, generate a prompt (automatic or the colour mask from section 3), predict masks, keep the best by IoU, clean up with morphological operations.
2. Align the two dates' masks. Erosion = land at t1 that is water at t2; accretion = the reverse. Patches under 500 pixels are dropped.
3. Area = eroded pixel count × pixel area (100 m² at 10 m).

Results

- Mean IoU 86.3%, Dice 92.6%; on unseen sites IoU rose from 0.66 (base SAM) to 0.87.
- Erosion area was accurate: 11.77 km² predicted vs 11.75 km² in the ground truth.
- Accretion was underestimated: 12.07 vs 14.12 km², so net change came out at +0.30 instead of +2.37 km².

Limits the paper lists: events under 0.1 ha are missed, 2–3 year gaps between images, about 8.3% misclassification from sediment, and the need for a 16 GB GPU.

Other SAM adaptations it cites: mariculture [31], cropland [13], and Erosion-SAM for soil erosion [33].

Relevance to a predictor: fine-tuned SAM is a strong way to generate erosion labels from imagery, but it only detects. Its output would be the target your predictor learns to forecast.

8. Future directions the paper proposes

These are suggestions, not yet tested in the paper:

Direction	What it means	How it would be used for erosion
Multi-sensor fusion	Combine Sentinel-1 radar, optical imagery and a DEM	Stack them as input channels so the model sees through monsoon clouds (radar), gets colour detail (optical) and bank height (DEM)

Direction	What it means	How it would be used for erosion
Temporal deep learning	LSTMs or Transformers that read a sequence of images	Feed several years of masks or features per location and predict next season's erosion, which turns detection into forecasting
Active learning	The model picks the images it is least sure about for a person to label	Cuts labelling effort, since only the most useful scenes get annotated
Larger models	ViT-G encoders, SAM 2 (which tracks objects across video frames)	SAM 2 could follow a bank line through an image time series
Bayesian uncertainty	Output a confidence with each prediction	Lets agencies act first on high-confidence erosion warnings
Physics-informed learning	Add hydraulic rules (flow, sediment, bank strength) to the model or its loss	Keeps predictions physically plausible where training data are thin
Explainable AI	Show which inputs drove a prediction	Builds trust with BWDB and local officials
Federated learning	Train across agencies without sharing raw data	CEGIS, BWDB and universities could pool models while keeping their data

The paper also proposes scaling to national-scale mapping and linking erosion to population and infrastructure data.

What this means for an erosion predictor

The paper is a detection study: it maps erosion after it has happened and does not forecast it. The only predictive approaches it mentions are the fuzzy susceptibility model [21], the ANN course forecast [24], and its suggestion of LSTMs or Transformers. Those are the references to follow for a predictor.

Sources

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